

Shanu Kumar

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LinkedIn | GitHub | Portfolio

SUMMARY

DevOps Engineer with 4+ years of experience designing, building and operating cloud-native infrastructure on AWS and Kubernetes. Experienced in building scalable CI/CD platforms, implementing Infrastructure as Code, and enabling reliable microservice deployments across large engineering organizations. Strong focus on platform engineering, site reliability practices, and infrastructure automation to improve system reliability, developer productivity, and operational efficiency.

TECHNICAL SKILLS

- **AWS:** EKS, EC2, RDS, S3, VPC, IAM, Lambda, DynamoDB, Route53, CloudFront
- **Containers & Platform:** Kubernetes, Docker, Helm, KEDA, Karpenter, Istio, Terraform, CDKTF, CloudFormation
- **CI/CD & GitOps:** Azure DevOps, GitHub Actions, GitLab CI, Jenkins, ArgoCD
- **Observability & Security:** Prometheus, Grafana, Loki, Datadog, Trivy, DependencyTrack
- **Programming:** Python, TypeScript, Bash

EXPERIENCE

Lead Engineer

Verisk Analytics

Apr, 2024 – Present

Hyderabad, India

- Led migration from Kubernetes Ingress to Gateway API, modernizing traffic management across platform clusters and enabling scalable routing policies for 100+ microservices.
- Architected and executed service mesh migration from sidecar to ambient mode using Istio, reducing service mesh resource consumption by 25% and eliminating per-pod proxy overhead.
- Built scalable CI/CD execution platform by migrating Azure-hosted runners to self-hosted runners on Amazon EKS clusters, reducing CI infrastructure costs by 30%.
- Led migration of legacy infrastructure provisioning to Terraform-based Infrastructure as Code, enabling version-controlled deployments and reducing configuration drift across environments.
- Implemented software supply chain security by integrating DependencyTrack into CI pipelines, enabling automated vulnerability scanning across 50+ services.
- Designed reusable Terraform modules and Kubernetes deployment templates for platform services such as Amazon RDS and internal platform components, reducing environment provisioning time by 90%.
- Provided platform engineering leadership through architecture reviews, production incident debugging, and mentoring engineers across multiple development teams.

Technical Lead

HCL Technologies

Oct, 2022 – Mar, 2024

Pune, India

- Designed and implemented cloud-agnostic CI/CD pipelines across Kubernetes and AWS environments, reducing release cycle time by 50% and improving deployment reliability.
- Built reusable infrastructure modules and application templates using Helm and Terraform, standardizing infrastructure provisioning and microservice deployments across teams.
- Owned lifecycle management of production Amazon EKS clusters including version upgrades, add-on compatibility, autoscaling configuration, and cluster observability.
- Enabled distributed data processing workloads by deploying Apache Spark on Kubernetes, supporting scalable analytics pipelines.

Senior Software Engineer

HCL Technologies

Aug, 2021 – Oct, 2022

Pune, India

- Automated AWS infrastructure provisioning using Terraform and AWS CloudFormation across development and production environments.
- Standardized Kubernetes microservice deployments using Helm and GitOps workflows with ArgoCD, improving deployment consistency across 50+ services.
- Implemented blue-green and canary deployment strategies, reducing production deployment risk and enabling safer application releases.
- Built CI/CD pipelines for containerized microservices, enabling automated build, test, and deployment workflows.

EDUCATION

Master of Technology in Computer Science

Indian Institute of Technology Madras

Jul 2019 – Jul 2021

Chennai, India

CERTIFICATIONS

- **Certified Kubernetes Administrator (CKA)**, 2023
- **HashiCorp Terraform Associate**, 2023
- **Certified Kubernetes Application Developer (CKAD)**, 2023
- **AWS Certified AI Practitioner**, 2026